

### generalised Fermat curve

$$C = X_{d,1}(\lambda) \subset \mathbb{P}^N$$

$C \rightarrow \mathbb{P}^1$  Galois, with group

$$H = \mu_d^{N+1} / \mu_d, \text{ of genus}$$

$$1 + \frac{1}{2}d^{N-1}((N-1)(d-1) - 2)$$



### the complete intersection

$$X \cong X_{d,r}(\lambda)$$

$c = N - r$  diagonal equations

$$\Phi(u)_i = \prod_j u_i^{(j)}, X \cong C^r / G$$

$$\deg \Phi = |G| = r! d^{N(r-1)}$$

$X \in \mathcal{A}$ : (F3') in every degree



### the Hodge classes of degree $r$

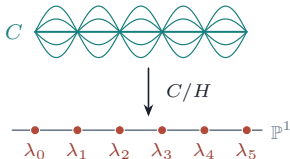
$$H^r(X) = \mathbb{C}\eta \oplus \bigoplus_{[a]} W_{[a]}$$

$$W_{[a]} = \bigoplus_{a' \in [a]} \bigwedge^r V_{a'}$$

$$\dim V_a = \#\{i : a_i \neq 0\} - 2$$

$W_{[a]}$  is the image of  $H^r(B_{[a]})$

under an algebraic correspondence



**known:**  $\dim B_{[a]} \leq 5$

Markman, Moonen–Zarhin

$d \in \{3, 4, 6\}$  and  $N \leq 6$ ,

or  $d = 2$  and  $N \leq 12$ ;

70, 490, 6125 Weil orbits in  $\mathbb{P}^6$

**open:**  $\dim B_{[a]} > 5$

$N \geq 7$  and  $d \in \{3, 4, 6\}$ ,

or  $N \geq 13$  and  $d = 2$

a case of (F2)

**open:**  $\varphi(e) \geq 4$

$d = 5$  or  $d \geq 7$

Weil classes of  $\mathbb{Q}(\zeta_e)$

P2<sub>split</sub> of the paper

**very general  $\lambda$ , every  $N$ :** the Hodge conjecture holds for  $d = 2$  in every dimension and for  $d = 3$  up to dimension 7;

for  $d \in \{4, 6\}$  and  $r \leq 4$  it follows from an analogue of the theorem of Achter and Pries for cyclic covers of degree 4 and 6